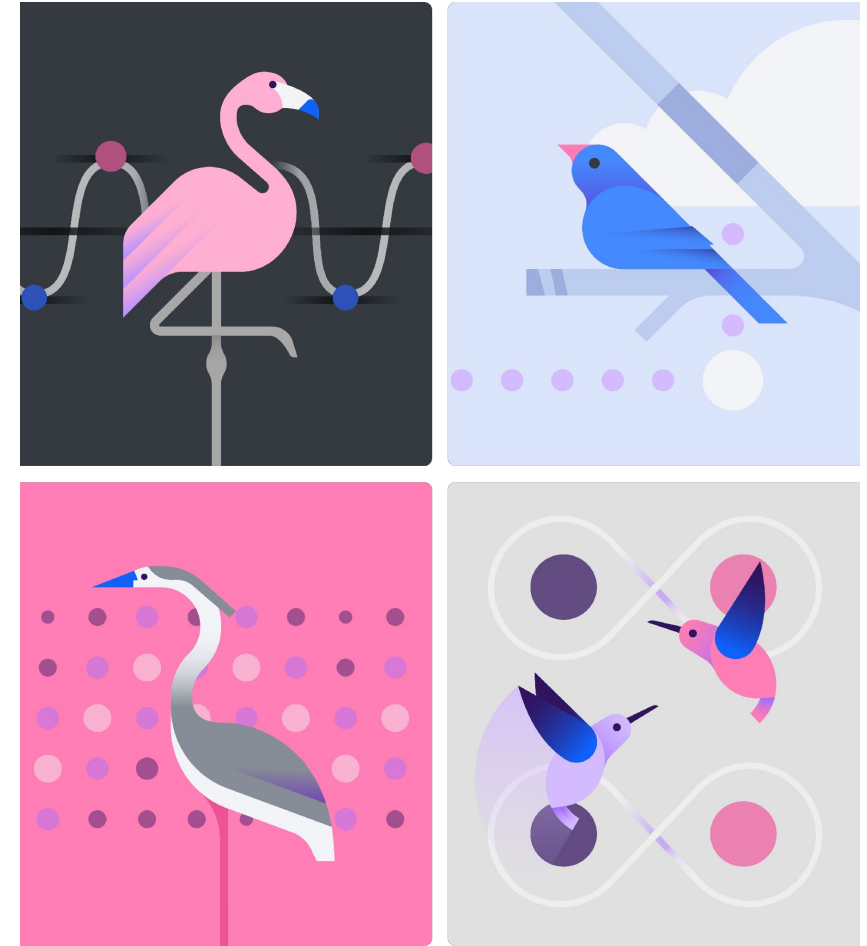


Lab 00

Getting started with Qiskit in Google Colab

Environment setup and first circuit

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What Lab 00 prepares you to do



01 SET UP

Install Qiskit in a fresh Google Colab runtime and verify the version.

02 BUILD

Create a two-qubit circuit using H, CNOT, and measurement.

03 EXECUTE

Run 1,024 simulated shots with StatevectorSampler.

04 EXPLAIN

Interpret Bell-state counts and test how each gate changes the result.



A quantum program needs more than Python

Python provides the language

Qiskit provides circuit objects, primitives, transpilation, and visualization.

A runtime provides the CPU, memory, packages, and temporary storage that execute the notebook.

Environment setup makes every later experiment reproducible.

Google Colab and a local Python environment



GOOGLE COLAB

- Browser based
- Python already available
- No local installation required
- Runtime resets after inactivity

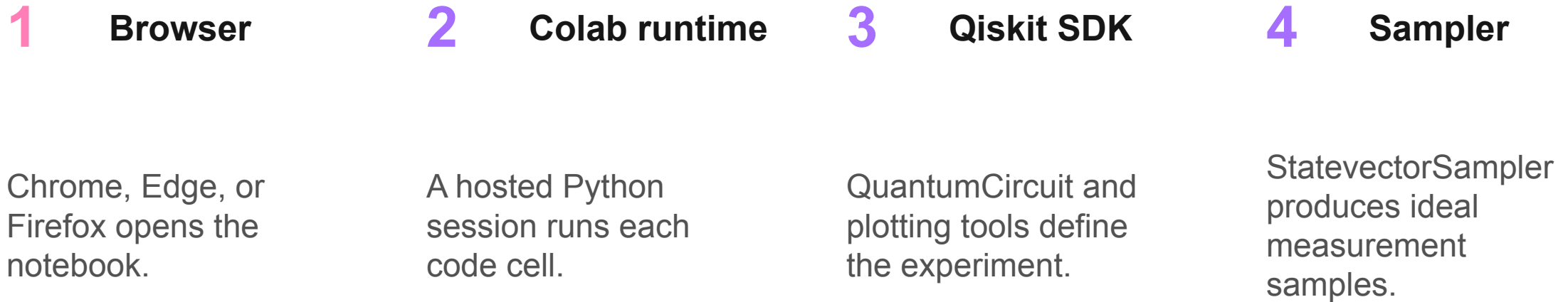
LOCAL COMPUTER

- Works offline after setup
- You control Python and package versions
- Files persist between sessions
- Requires installation and maintenance

Course default: Google Colab



The Lab 00 software stack



Lab 03 later replaces the local sampler with IBM Quantum hardware.

Installing Qiskit in Colab



```
%pip install -q "qiskit[visualization]"
```

%pip installs into the active notebook environment

[visualization] adds packages for circuit drawings and plots

-q reduces installation messages

Run this cell whenever Colab starts a new runtime.



Verification before circuit construction

CODE

```
import qiskit

print("Qiskit version:",
      qiskit.__version__)
```

SUCCESS

No import error
A version number appears
The runtime is ready

Notebook reference version

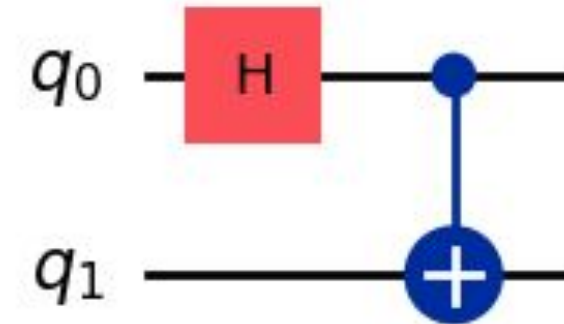
Qiskit 2.5.2

Record the version when sharing results or asking for help.

The first two-qubit circuit

```
qc = QuantumCircuit(2)

qc.h(0)
qc.cx(0, 1)
```



Initial state

$|00\rangle$

After H and CNOT

$(|00\rangle + |11\rangle) / \sqrt{2}$



H and CNOT play different roles

HADAMARD ON q_0

Creates equal amplitudes for $q_0 = 0$ and $q_0 = 1$.

The next measurement becomes probabilistic.

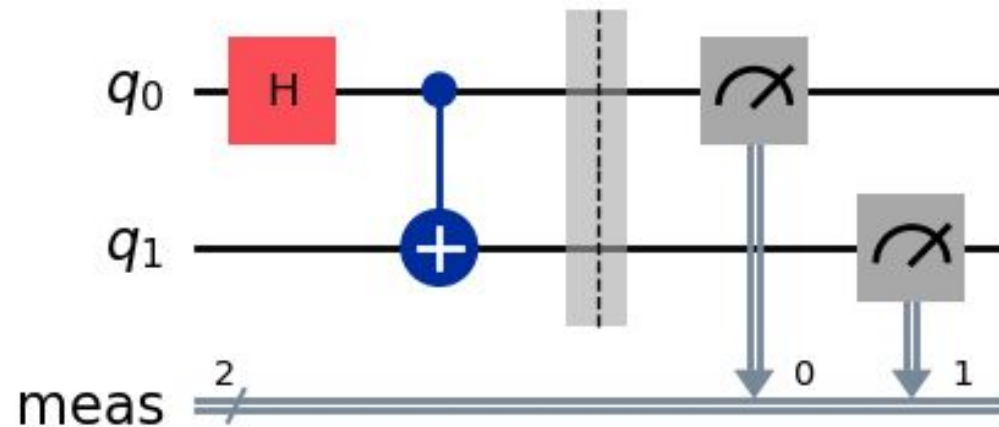
CNOT q_0, q_1

Flips q_1 only when q_0 is 1.

Applied after H, it creates correlation and entanglement.

CNOT alone leaves $|00\rangle$ unchanged because the control qubit equals 0.

Measurement converts qubits into classical results



`qc.measure_all()`

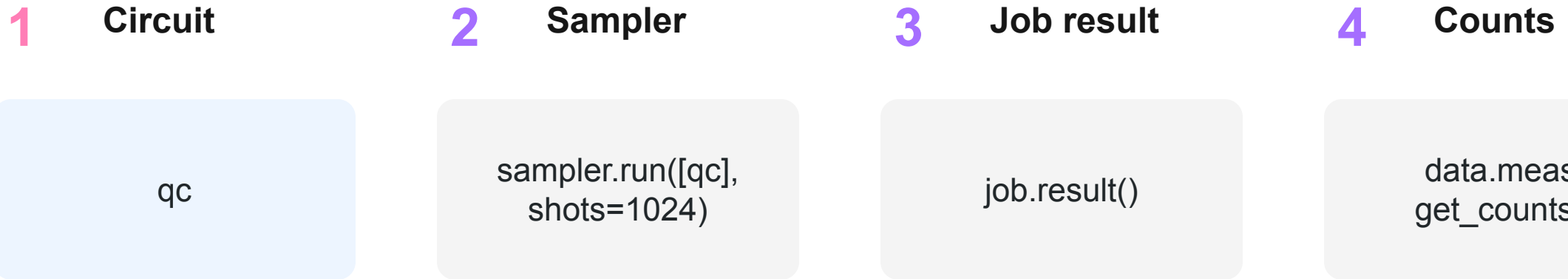
adds measurement operations and
classical bits

One shot returns one bit string

1,024 shots estimate a distribution

Expected Bell outcomes: 00 and 11

StatevectorSampler execution workflow



Simulation provides an ideal reference before the hardware labs introduce noise.

Experiments in the Lab 00 notebook

01	Remove H	CNOT receives control 0, so the result remains 00.
02	Remove CNOT	q0 varies but q1 remains 0, so the qubits are not entangled.
03	Change the shots	Observed frequencies stabilize near the theoretical 50% and 50%.
04	Apply X before CNOT	X prepares control 1, so CNOT produces 11 deterministically.

Ready for Lab 00



Before opening the notebook

- Google account available
- Colab opens in the browser
- Predictions recorded before execution
- IBM Quantum account not required today

Predict
Build
Execute
Observe
Explain

Next: run the installation cell and verify the Qiskit version.